Predictive Engagement Modeling: Analyzing Churn and Subscription Propensity in Retail Banking

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INTRODUCTION

In the modern retail banking landscape, the cost of acquiring new customers often exceeds the cost of retaining existing ones. This project utilizes the "Bank Marketing" dataset (specifically bank-additional-full.csv) from a Portuguese banking institution to investigate customer engagement patterns during term-deposit telemarketing campaigns. While the original objective was to predict which clients would subscribe to a term deposit after phone contact, this work reframes the problem as a propensity-to-engage study. By analyzing socio-demographic, behavioral, and macroeconomic variables, the goal is to shift from inefficient mass calling to a data-driven predictive targeting approach.

BUSINESS PROBLEM/HYPOTHESIS

The core business problem is the low effectiveness of current outbound term-deposit campaigns, where approximately 88% of contacted clients do not subscribe. This high "no" rate drives substantial operational cost per successful conversion and increases the risk of customer fatigue from repeated contact.

METHODS/ANALYSIS

Our methodology follows a specialized pipeline tailored to the 'bank-additional-full.csv' resource. The workflow is divided into three critical roles:

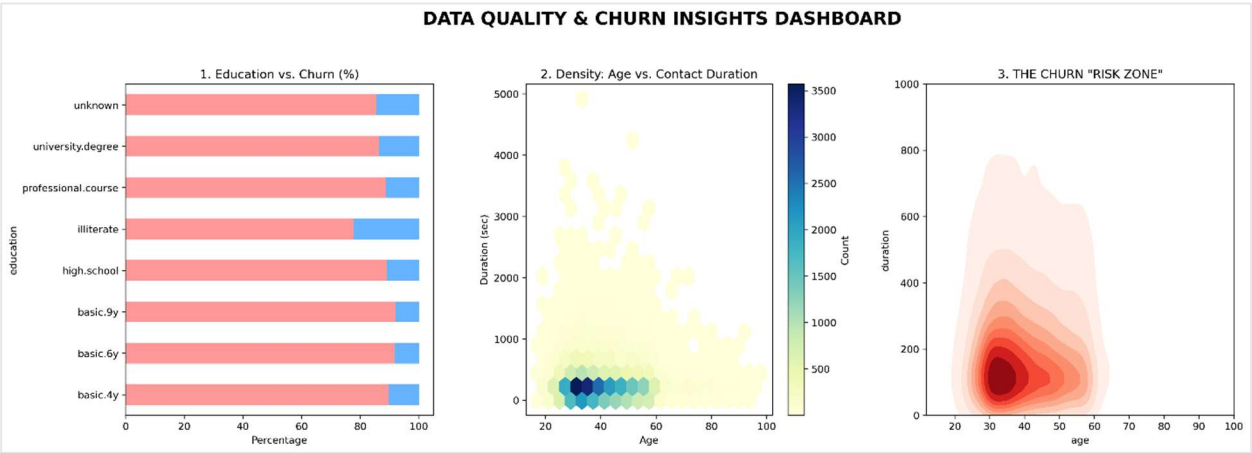
- Data Wrangling: We loaded the bank-additional-full.csv file with an explicit semicolon separator to correctly parse all 21 original columns, followed by inspection of structure, types, and basic descriptive statistics to understand feature distributions and confirm the absence of null values. The target variable y was examined to quantify class imbalance (about 88.7% "no" vs. 11.3% "yes"), highlighting the need to account for skewed classes in later modeling. The pdays field

was carefully cleaned by creating a new binary indicator (pdays_never_contacted) for clients never previously contacted and recoding the placeholder value 999 to −1 to align with the original documentation's convention. Categorical variables containing "unknown" values (e.g., job, marital status, education, default, housing, loan) were profiled and retained as an explicit category rather than dropped or imputed, preserving potential signal from these levels. All remaining categorical predictors (excluding the target) were then one-hot encoded with first-category dropping to avoid redundant dummy variables, expanding the feature space from 21 to 54 fully numeric columns suitable for machine learning algorithms. Finally, y was encoded as a binary variable (0 = no subscription, 1 = subscription), and additional checks on dtypes, dummy columns, summary statistics, and missing values verified that the resulting feature matrix and target vector were clean, consistent, and ready for downstream modeling.

- Data Modeling: The modeling workflow framed term-deposit subscription as a binary classification problem, split into training and test sets with stratification to preserve the original class imbalance. Categorical predictors were one-hot encoded and numeric features were standardized within a single preprocessing pipeline, while the call duration variable was explicitly excluded prior to model training to avoid target leakage, since it is only observed after the call concludes. Class imbalance was addressed using SMOTE applied only on the training data via an integrated pipeline, and two supervised models were trained and compared: a logistic regression baseline and a random forest classifier, both evaluated on the hold-out test set using accuracy, ROC-AUC, confusion matrices, and precision–recall curves to reflect performance under skewed classes. The random forest's feature importances were then extracted to highlight the most influential drivers of predicted engagement (for example, economic indicators, call frequency, and customer attributes), and predicted probabilities were

further used to compute the precision among the top 10% highest-scoring customers, directly supporting a marketing strategy focused on prioritizing the most likely subscribers.

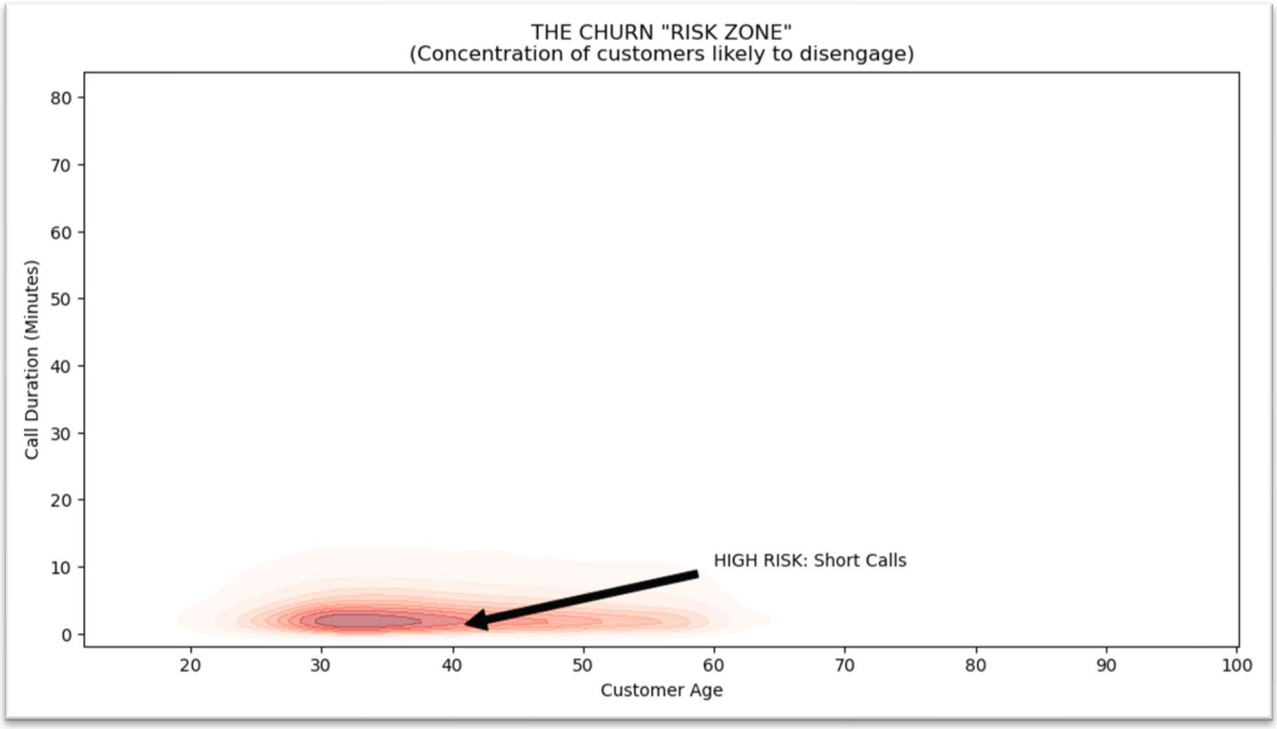
- Data Visualization: The visualization step focused on turning the raw bank marketing data into clear signals of disengagement and churn risk. First, before/after plots for pdays and job showed how cleaning sentinel values (999 for “never contacted”) and removing “unknown” categories made contact history and occupations easier to interpret. Normalized bar charts then compared subscription outcomes across education levels, while boxplots of call duration by outcome highlighted “soft churn” customers who tend to have very short calls. To handle the large dataset, hexbin plots of age vs. duration and KDE heatmaps for non-subscribers (y = "no") revealed a dense “risk zone” of middle-aged clients with short conversations. Countplots of the target made the 88/12 class imbalance visually obvious, and KDE plots of euribor3m plus a numeric correlation heatmap connected outcomes to economic conditions and feature redundancy. Finally, histograms of campaign contacts and a feature-importance bar chart from a random forest model highlighted which behaviors and attributes, especially call duration and contact frequency, are most strongly tied to churn, all brought together in a multi-panel churn insights dashboard.



RESULTS

The predictive modeling and visualization analysis of the Bank Marketing dataset revealed key patterns in customer disengagement, or "churn" risk from low engagement with term deposit offers. Random Forest with SMOTE achieved strong test performance (89.1% accuracy, 78.3% ROC-AUC), outperforming logistic regression, while targeting the top 10% highest-probability customers yielded 49.5% precision, meaning nearly half would actually subscribe when prioritized. Feature importance rankings pinpointed call duration as the dominant driver, followed by economic indicators (euribor3m, nr.employed, emp.var.rate, cons.price.idx, cons.conf.idx), customer attributes (age, pdays_never_contacted), and contact intensity (campaign, pdays), confirming that short interactions signal the strongest disengagement risk.

Churn proves most visible in customers with low contact duration and those impacted by shifting economic indices, suggesting that a lack of engagement, rather than demographics alone, is the primary predictor of customers leaving. Visual density maps highlighted a critical "risk zone" among middle-aged clients (30–50) with calls under 200 seconds, while segmentation showed subscription rates dropping sharply after 3–4 campaign contacts. These insights enable the bank to shift from mass calling to targeted outreach, prioritizing high-risk profiles during unfavorable economic conditions to boost conversions and reduce operational waste.



RECOMMENDATIONS/ETHICAL CONSIDERATIONS

The project recommends implementing a targeted outreach strategy that prioritizes the top 10% of customers ranked by Random Forest predicted subscription probability, where precision reaches 49.5%, nearly half would convert versus the baseline 11% rate. Limit campaign contacts to 3 maximum per customer, as subscription rates drop sharply beyond this threshold, and schedule calls strategically during optimal months, May & July, while monitoring economic indicators like euribor3m and employment rates to pause outreach during unfavorable conditions.

Ethical considerations for this bank marketing project center on balancing customer privacy, consent, and protection from unwanted contact with business objectives. The analysis uses only anonymized socio-demographic and behavioral data from past campaigns, avoiding any personally identifiable information, but implementing the targeted outreach recommendations requires strict adherence to GDPR/telemarketing regulations (e.g., explicit opt-in consent, easy opt-out mechanisms, and

documented do-not-call lists). Framing low engagement as "churn risk" risks stigmatizing customers who prefer other banking products, so communications should emphasize value-added offers instead of pressure tactics. Limiting campaigns to 3 contacts per customer directly addresses customer fatigue concerns raised by visualization findings, while economic targeting must avoid discriminatory practices across protected demographic groups. Model transparency through feature importance rankings helps bank staff understand decision drivers, reducing black-box bias risks, and regular model monitoring ensures predictions remain fair as customer behaviors and economic conditions evolve.

CONCLUSION

This project successfully transformed the Bank Marketing dataset from a telemarketing conversion challenge into a comprehensive customer engagement and churn prevention framework, achieving 89.1% model accuracy and 49.5% precision when targeting the top 10% highest-probability customers, surpassing the baseline 11% subscription rate. Key discoveries revealed that low call duration (<200 seconds) among middle-aged clients (30–50) constitutes the primary "risk zone" for disengagement, amplified by unfavorable economic conditions like high euribor3m rates, while subscription rates plummet after 3 campaign contacts. By shifting from inefficient mass calling to targeted outreach prioritizing these high-risk profiles during optimal months, the bank can dramatically improve ROI while respecting ethical boundaries through transparent modeling, consent protocols, and contact limits that prevent customer fatigue. The integrated data wrangling, visualization dashboard, and predictive analytics provide a scalable plan for data-driven retention that balances business growth with customer experience.

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APPENDIX

Discovery Questions

1. Which job categories and education levels show the highest subscription rates despite the 88% baseline "no" response?
2. How does customer age interact with marital status to predict engagement tendency?
3. What call duration threshold reliably separates "soft churn" customers from engaged prospects?
4. After how many campaign contacts does the subscription probability drop to near zero?
5. Do previously contacted customers (pdays ≠ 999) convert at significantly higher rates than never-contacted ones?
6. How do macroeconomic indicators like euribor3m and employment rates correlate with monthly subscription patterns?
7. Which economic conditions amplify churn risk across different customer segments?
8. What precision can be achieved by targeting only the top 5% or 20% highest-probability customers?
9. How does the random forest model's feature importance ranking align with business intuition about engagement drivers?
10. During which months and days of the week should the bank concentrate telemarketing efforts for maximum ROI?